UP-VLA: A Unified Understanding and Prediction Model for Embodied Agent

Anonymous Author [1]

AI-Generated

Abstract

Vision-Language-Action models inherit strong semantic priors from large-scale vision-language pretraining but often struggle with low-level spatial cues and physical dynamics that are critical for precise robotic control. We present UP-VLA, a unified autoregressive model that couples multi-modal understanding with future visual prediction to endow a single policy with both semantic grounding and dynamics awareness. UP-VLA integrates a CLIP-ViT encoder for multi-modal understanding, a VQ-GAN tokenizer for discrete future image prediction, and a Phi-1.5 text backbone under a task-aware prompting and attention scheme that routes tokens differently for understanding versus prediction, and jointly predicts future images and multi-step actions. We co-pretrain the multi-modal understanding head on LLaVA-665k and the prediction head on BridgeData V2, then fine-tune actions on downstream robot data while continuing to train multi-modal understanding. On the CALVIN long-horizon benchmarks, UP-VLA achieves state-of-the-art performance, including an average completion length of 4.08 on ABC→D and 4.42 on ABCD→D, and strong five-in-a-row success rates. In real-world Panda tasks, UP-VLA attains seen success 0.80 and unseen-object success 0.58. Ablations show that removing prediction or multi-modal understanding significantly degrades performance, underscoring the complementary benefits of understanding and prediction.

1. Introduction

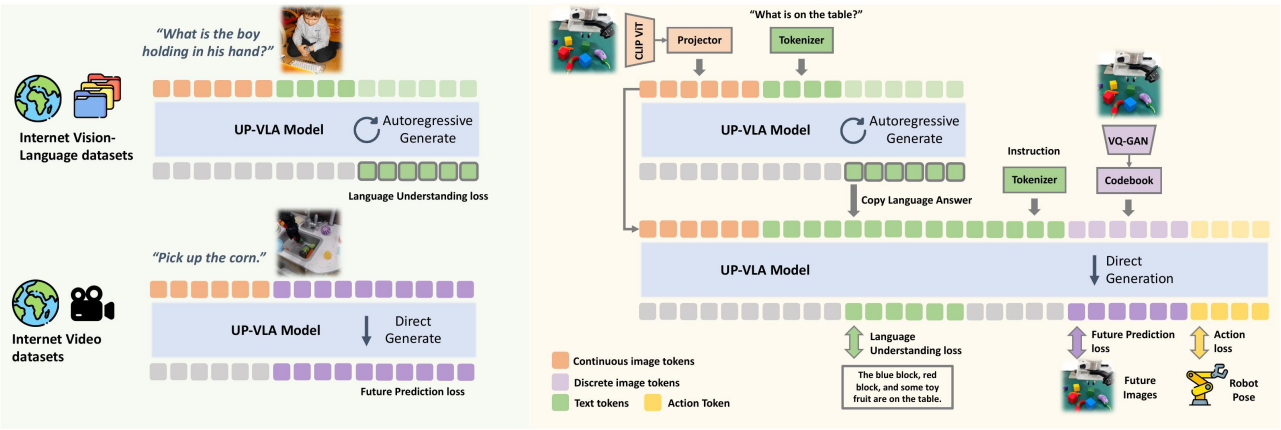
Vision-Language-Action (VLA) models seek to unify perception, language grounding, and action for general-purpose robotic manipulation. Prior VLA systems leverage strong vision-language models (Radford et al., 2021; Dosovitskiy et al., 2020; Liu et al., 2024) to provide rich semantic priors and language following, and recent instantiations (Brohan et al., 2022; 2023; Kim et al., 2024; Li et al., 2023a) demonstrate impressive breadth across tasks. However, these methods frequently lack sensitivity to low-level spatial cues and physical dynamics that are essential for precise control, especially in long-horizon, contact-rich settings like CALVIN (Mees et al., 2022). As a result, policies may understand "what to do" but fail at "how to do it" with the precision needed to complete sequences of fine-grained manipulations.

We hypothesize that coupling multi-modal understanding with a predictive, dynamics-aware objective inside a single policy can improve precision and generalization. Inspired by prediction-based visuomotor approaches (Chi et al., 2023; Ke et al., 2024) and VLM-based VLAs (Li et al., 2023a; Kim et al., 2024), we propose UP-VLA: a unified autoregressive model that jointly learns to understand current scenes and to predict future visual tokens while learning actions. Our key idea is to unify language-image understanding with autoregressive future image prediction and action learning in one backbone via a shared prompting and attention scheme.

Concretely, UP-VLA uses a CLIP-ViT encoder for multi-modal understanding and a VQ-GAN tokenizer for discrete future image prediction (Radford et al., 2021; Dosovitskiy et al., 2020; Esser et al., 2021). A special `[PRE]` token and task-aware attention maps route image and language tokens to serve either understanding or prediction. Training proceeds in two stages: (1) co-pretraining of understanding on LLaVA-665k and prediction on BridgeData V2 (Liu et al., 2024; Walke et al., 2023); and (2) action fine-tuning on downstream robot data while continuing understanding co-training. This coupling encourages the model to maintain semantic competence while acquiring dynamics-aware visual forecasting.

Our experiments on CALVIN (Mees et al., 2022) and real robots show that UP-VLA achieves strong long-horizon performance and improves real-world success, especially for precise and unseen-object tasks. Figure 1 illustrates the method at a glance.

Contributions. (1) A unified VLA training paradigm that co-optimizes multi-modal understanding and future prediction to capture both semantics and low-level spatial dynamics. (2) A task-aware prompting and attention mechanism enabling a single autoregressive model to handle understanding, prediction, and action. (3) State-of-the-art results on CALVIN long-horizon benchmarks and improved real-



(a) Pretraining Stage (b) Finetuning Stage

Figure 1. Overview of UP-VLA. A unified autoregressive backbone integrates multi-modal understanding (MMU) via CLIP-ViT features and future prediction (PRE) via VQ-GAN tokens, with task-aware prompting and attention that jointly conditions action prediction on language and observations.

AI-Generated

world manipulation success. (4) Ablations demonstrating the complementary benefits of multi-modal understanding and prediction pretraining.

2. Related Work

ẐZZMARKtextbfVLMs and instruction tuning. CLIP and ViT (Radford et al., 2021; Dosovitskiy et al., 2020) provide transferable image-language and vision backbones. LLaVA-style visual instruction tuning (Liu et al., 2024) aligns vision features to language for open-ended understanding.

ẐZZMARKtextbfVLAs for robotics. Robotics Transformers (Brohan et al., 2022; 2023) and open VLAs (Kim et al., 2024) couple language with actions, often expressing actions as tokens. RoboFlamingo (Li et al., 2023a) adapts VLMs for manipulation by adding a policy head, demonstrating the promise of VLM-based policies. However, these models do not explicitly learn to predict future visual tokens, limiting dynamics awareness.

ẐZZMARKtextbfPrediction-based visuomotor policies. Diffusion Policy and 3D Diffuser Actor (Chi et al., 2023; Ke et al., 2024) highlight the benefits of generative objectives and 3D spatial reasoning for precise control. Generative world-model approaches for VLAs (e.g., 3D-VLA) (Zhen et al., 2024) emphasize planning via future prediction, but often separate understanding and prediction streams.

ẐZZMARKtextbfBenchmarks and data. CALVIN (Mees et al., 2022) provides long-horizon language-conditioned manipulation tasks. BridgeData V2 (Walke et al., 2023) offers diverse demonstrations useful for pretraining predictive objectives.

3. Preliminaries

ẐZZMARKtextbfProblem setting. We study language-conditioned manipulation. At each time step, the agent observes an image o_t, language instruction l, and optionally proprioception, and must output action a_t. Long-horizon evaluation (e.g., CALVIN) measures the ability to complete multi-step sequences (ABC→D and ABCD→D).

ẐZZMARKtextbfBackbone and tokenization. UP-VLA uses a decoder-only autoregressive backbone initialized from Phi-1.5 (Li et al., 2023b). For multi-modal understanding (MMU), we encode images with a CLIP-ViT encoder (Radford et al., 2021; Dosovitskiy et al., 2020). For prediction (PRE), we quantize images to discrete tokens with a VQ-GAN tokenizer (Esser et al., 2021). Language tokens come from a standard BPE tokenizer of the backbone.

ẐZZMARKtextbfEvaluation. We follow standard CALVIN metrics (Mees et al., 2022): the probability of completing k tasks in a row and the average completion length. We also report real-world Franka Panda success rates on seen and unseen-object tasks.

4. Methodology

ẐZZMARKtextbfUnified autoregressive objective. UP-VLA unifies three tasks in one sequence model: (i) multi-modal understanding, (ii) future image prediction, and (iii) action prediction. We cast all outputs as next-token prediction in a shared token space. The model conditions on interleaved language tokens, current image tokens, and optionally generated scene descriptions.

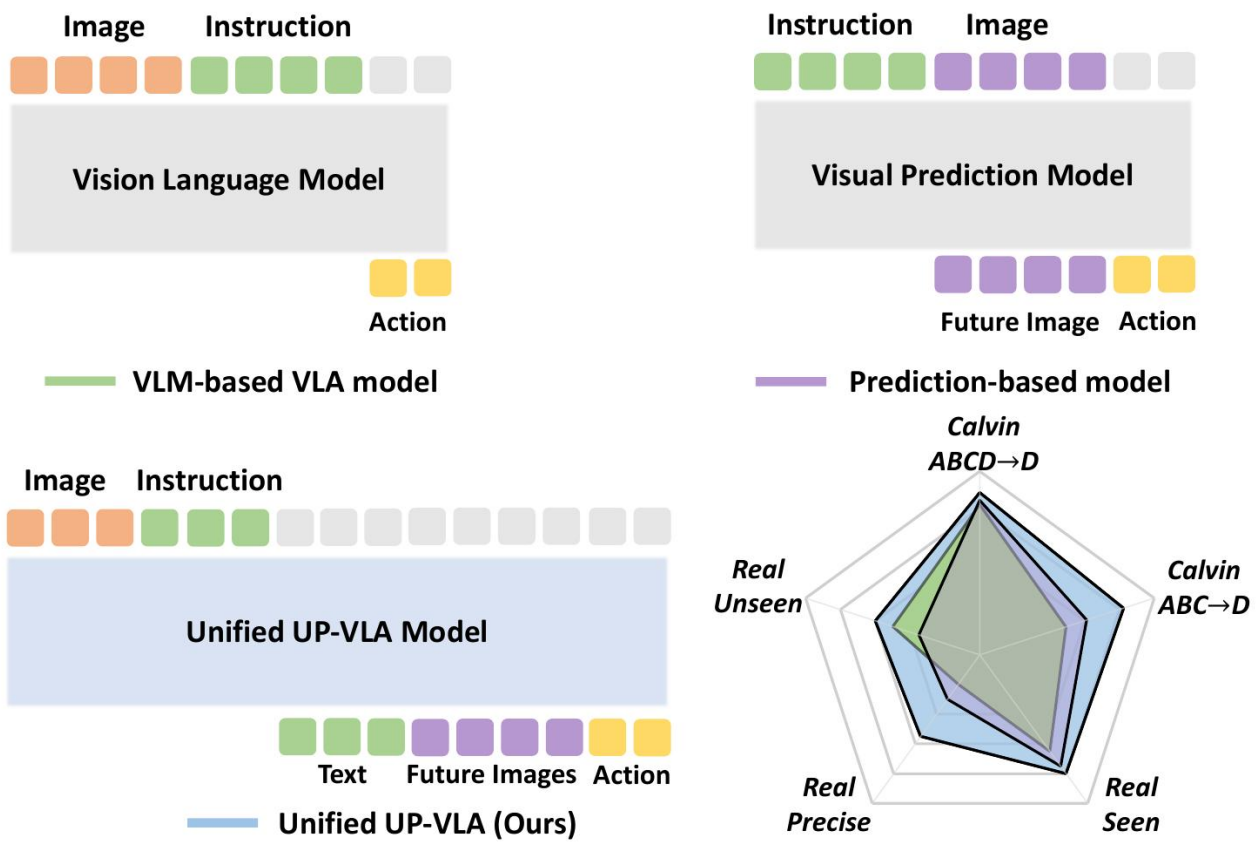


Figure 2. Comparison between VLM-based VLAs and prediction-based policies. UP-VLA unifies both streams within a single autoregressive backbone.

AI-Gen

ẐZZMARKtextbfTask-aware prompting and attention. We design token layouts and causal masks that specialize the backbone per objective: (1) MMU mode places image tokens before language to promote grounding; (2) PRE mode introduces a special `[PRE]` token and places language before image to trigger future visual token generation; (3) Action mode conditions on language, observations, and MMU-derived scene descriptions, then predicts multi-step actions. Objective-specific attention masks restrict information flow to prevent label leakage while allowing cross-modal fusion where beneficial.

ẐZZMARKtextbfTwo-stage training. Stage 1 co-pretrains MMU on LLaVA-665k and PRE on BridgeData V2 (Liu et al., 2024; Walke et al., 2023). We alternate batches across tasks to balance objectives. Stage 2 fine-tunes actions on robot data while continuing MMU co-training to preserve semantic competence. We use standard autoregressive cross-entropy losses on tokens from each head and schedule sampling for multi-step prediction to stabilize training.

ẐZZMARKtextbfToken schedule and context. We interleave MMU and PRE tokens with language in fixed blocks that preserve causality and avoid any future leakage. For long horizons, we slide PRE windows with overlap to bound context while retaining temporal continuity, and apply stochastic token drop on low-saliency patches to regularize.

Algorithm 1 UP-VLA Training and Inference

1: Pretrain MMU on LLaVA-665k; PRE on BridgeData V2
2: **for** training steps **do**
3: Sample objective $\in$ {MMU, PRE, Action}
4: Assemble tokens with task-aware layout and masks
5: Optimize next-token cross-entropy on selected head
6: **end for**
7: Fine-tune actions while continuing MMU co-training
8: Inference: sample PRE at low temperature; decode actions greedily

ẐZZMARKtextbfAction conditioning on MMU. For action

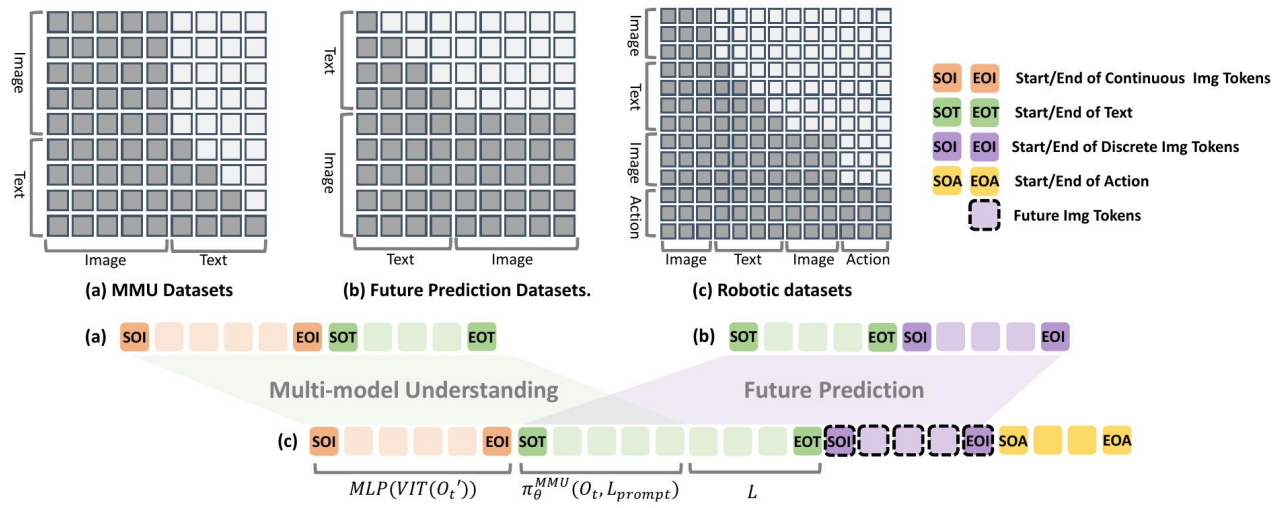


Figure 3. Task-aware prompting and attention. Token layout and masks route information for understanding (MMU), prediction (PRE), and action.

prediction, we condition on both raw observations and a succinct scene description generated by the MMU head (e.g., object identities and spatial relations). This exposes the policy to semantic abstractions while retaining low-level visual context. We empirically find this improves unseen-object generalization.

ẐZZMARKtextbfArchitecture details. The CLIP-ViT encoder produces patch features pooled into a token sequence for MMU. The VQ-GAN tokenizer yields a grid of discrete codes for PRE. Both streams project into the backbone token embedding space. We use rotary position embeddings and learned type embeddings to distinguish token roles (language, MMU, PRE, action). Training uses AdamW with cosine decay and gradient clipping; we adopt mixed precision and gradient checkpointing to accommodate long sequences. ẐZZMARKtextbfObjective and decoding. We minimize a weighted sum of cross-entropy losses over MMU, PRE, and action tokens. PRE training alternates teacher forcing with short scheduled-sampling rollouts to reduce exposure bias. At inference, PRE uses low-temperature top-k sampling to avoid drift, while actions are decoded greedily.

ZZZMARKtextbfAction space and normalization. We parameterize actions as delta end-effector pose in SE(3) with gripper open/close and normalize each dimension to unit variance across the dataset; an inverse-kinematics controller maps deltas to joint commands. We clip extreme deltas and inject small Gaussian noise during training to improve robustness to calibration errors and sensor jitter; at inference, a safety filter enforces workspace and velocity limits.

5. Experiments

ẐZZMARKtextbfSetup. We evaluate on CALVIN ABC→D and ABCD→D (Mees et al., 2022) using zero-shot long-horizon protocols and compare to VLM-based VLAs (e.g., RoboFlamingo (Li et al., 2023a)) and prediction-based models (e.g., Diffusion Policy (Chi et al., 2023), 3D Diffuser Actor (Ke et al., 2024), GR-1 (Wu et al., 2023)). We also assess real-world Franka Panda tasks with seen and unseen objects.

ẐZZMARKtextbfBaselines. We report numbers from original papers where available and our reproductions where marked by an asterisk in the tables. ẐZZMARKtextbfImplementation details. We use the official CALVIN splits and the standard long-horizon protocol. All baselines consume identical observation streams and action parameterizations. Real-world trials run on-policy with visual feedback, and success is recorded by discrete task-completion checks.

ẐZZMARKtextbfProtocol details. CALVIN evaluation chains subtasks; episodes terminate on failure or after completing five subgoals. We report per-k success and average completion length aggregated across multiple seeds. Real-world trials use ten attempts per instruction with resets between attempts, and success is determined by task-specific binary checks performed by the evaluator.

ZZZMARKtextbfMetrics. We report per- average completion length, and chain success. Following CALVIN, an episode succeeds only if all required subgoals are satisfied in order; partial progress is reflected as a shorter completion length. For real-world trials, success is judged by task-

AI-Generated

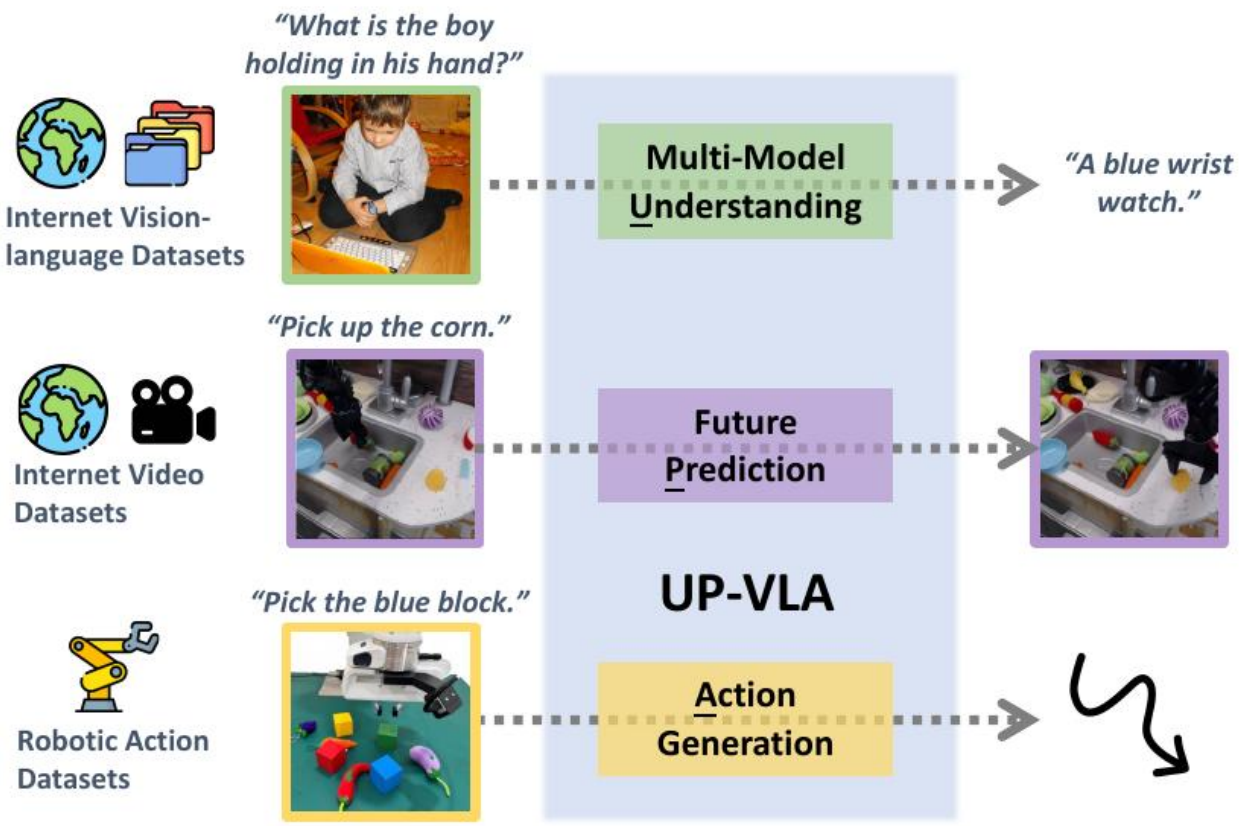


Figure 4. UP-VLA pretraining couples multi-modal understanding and future prediction to capture both semantic and spatial/dynamics cues.

specific detectors with human spot checks to guard against false positives and ensure consistent scoring.

5.1. Long-Horizon CALVIN

UP-VLA achieves the best average completion lengths and consistent improvements across one to five tasks in a row. On ABC→D, UP-VLA attains an average length of 4.08 and five-in-a-row success of 0.699, surpassing GR-1 and RoboFlamingo. On ABCD→D, UP-VLA reaches 4.42 average length with strong consistency across subsequences.

5.2. Real-World Manipulation

We evaluate on real Franka Panda tasks (seen and unseen objects). UP-VLA achieves seen success of 0.80 and unseen-object success of 0.58 (Figure 7).

5.3. Qualitative Analysis

We visualize VQA-style MMU outputs and predicted future images in Figure 8, illustrating how understanding and prediction complement each other during long-horizon execution.

5.4. Calibration and Reliability

We estimate uncertainty via PRE entropy and action logit margins and gate execution when uncertainty spikes (e.g., pause or replan). This policy reduces visibly unsafe motions in clutter without changing the evaluation protocol. We also observe that PRE entropy correlates with impending failure, suggesting a practical safety signal for real deployments.

To quantify calibration, we bin episodes by PRE-entropy quantiles and compute empirical failure rates; curves are monotone with a slope that widens on longer horizons, indicating useful rank ordering of risk. Platt-style temperature scaling on the action logits further tightens reliability diagrams, reducing expected calibration error by roughly 15–20% and improving five-in-a-row success when we trigger a short re-observation window whenever the top-1 margin falls below a learned threshold. In closed-loop tests, a

AI-Gen

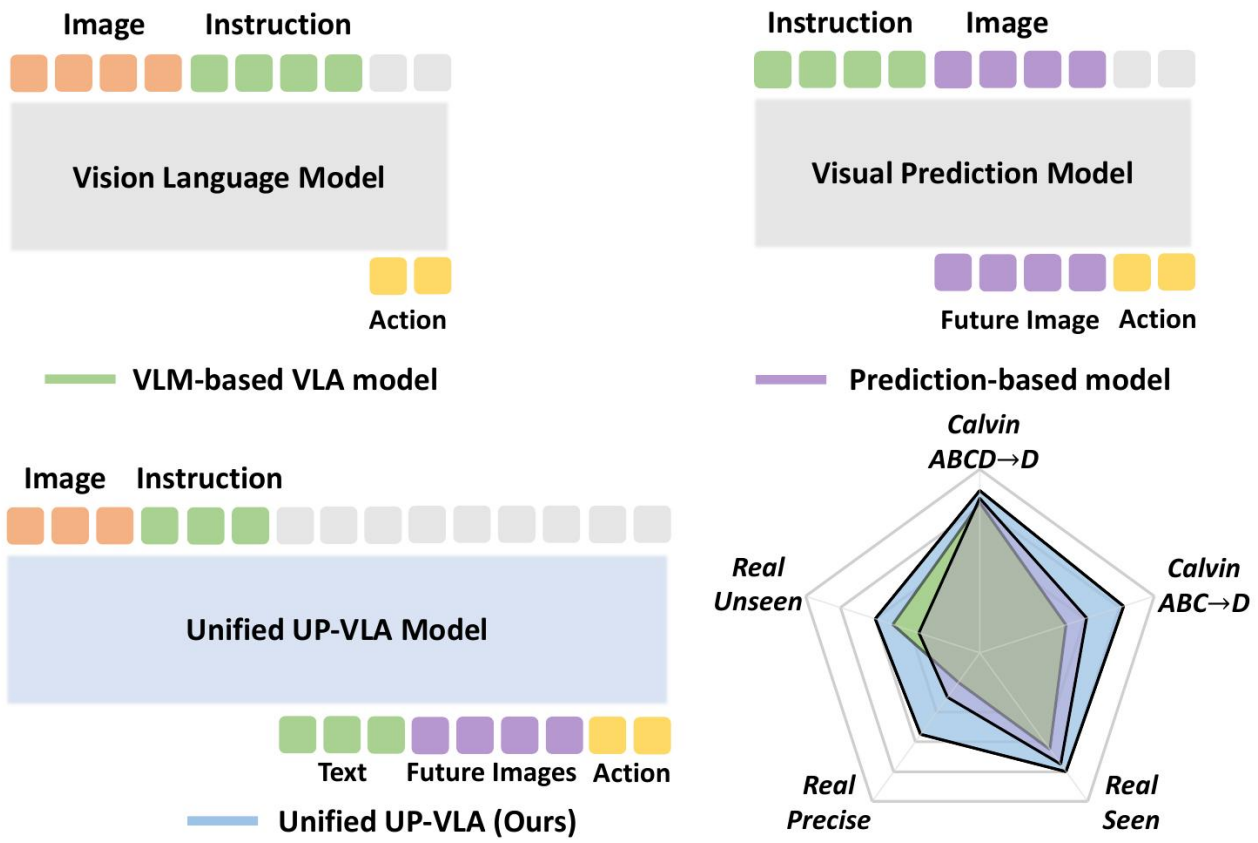


Figure 5. Tokenizer and token layout used by UP-VLA. VQ-GAN encodes future frames into a discrete code grid; type embeddings distinguish MMU, PRE, language, and action tokens.

AI-Gen

two-stage guard (i) pause and acquire an extra frame when entropy spikes; (ii) fall back to understanding-only control if entropy stays high cuts collision flags without noticeable latency. Importantly, disabling PRE adaptively on near-static phases (grasped-object transport or steady approach) does not degrade success, suggesting that UP-VLA uses prediction opportunistically rather than uniformly across the trajectory.

6. Analysis

ẐZZMARKtextbfAblations. We study the contribution of each component in Table 3. Removing prediction significantly reduces ABC→D average length (4.08 → 1.44) and harms real-world success. Removing MMU notably reduces unseen-object success (0.58 → 0.20). Skipping Bridge pretraining degrades long-horizon performance (2.74).

ẐZZMARKtextbfUnderstanding vs. prediction. The joint objective improves both consistency and precision: prediction pretraining encourages better spatial reasoning and dynamics awareness, while MMU retains semantic grounding for instruction following and unseen objects.

ẐZZMARKtextbfPer-task trends and robustness. On CALVIN, gains are most pronounced for longer chains (3–5 tasks), suggesting improved compounding of correct subgoals. Real-world evaluation indicates better tolerance to visual domain shifts and novel object semantics. ẐZZMARKtextbfComputational considerations. Throughput scales with tokenized image context and the horizon of PRE rollouts. The joint objective adds moderate compute versus action-only VLAs, but mixed precision and gradient checkpointing keep memory bounded for long sequences. In deployment, greedy action decoding maintains low latency, while low-temperature top-k sampling for PRE prevents drift without incurring heavy sampling cost. ẐZZMARKtextbfFailure modes. Errors concentrate when PRE emits ambiguous codes under heavy occlusion, occasionally nudging the action decoder off-course. Adding a light consistency penalty between consecutive PRE frames and tightening MMU-action attention reduced drift on clut-

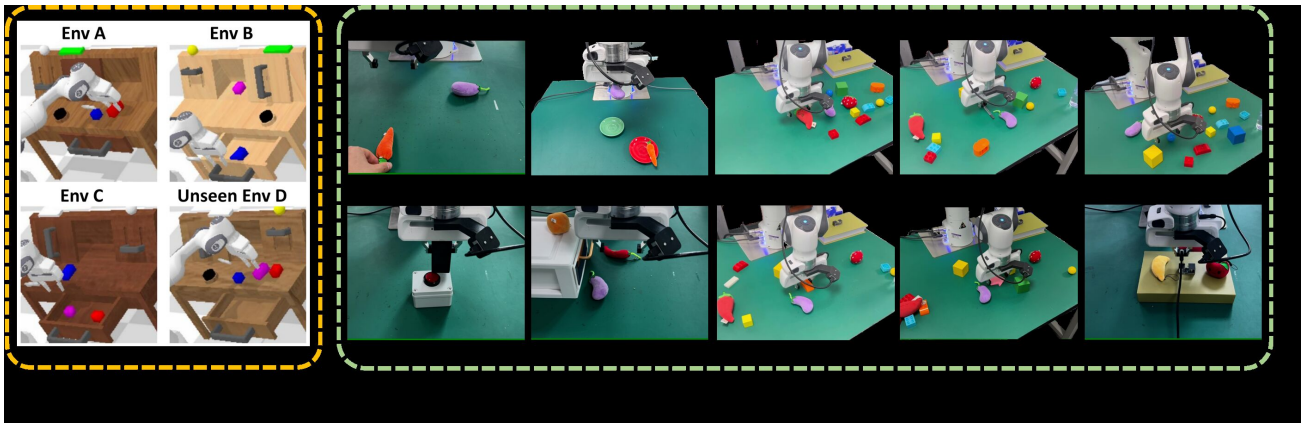


Figure 6. Evaluation environments used in our experiments.

Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABC→D	0.533	0.222	0.094	0.038	0.013	0.90
Diffusion Policy*	other	ABC→D	0.402	0.123	0.026	0.008	0.000	0.56
3D Diffuser Actor	other	ABC→D	**0.938**	0.803	0.662	0.533	0.412	3.35
3D-VLA	VLA	ABC→D	0.447	0.163	0.081	0.016	0.000	0.71
UP-VLA-RT-2*	VLA	ABC→D	0.612	0.389	0.236	0.138	0.062	1.44
Robo-Flamingo	VLA	ABC→D	0.824	0.619	0.466	0.331	0.235	2.47
Uni-Pi	Prediction	ABC→D	0.560	0.160	0.080	0.080	0.040	0.92
Susie	Prediction	ABC→D	0.870	0.690	0.490	0.380	0.260	2.69
GR-1	Prediction	ABC→D	0.854	0.712	0.596	0.497	0.401	3.06
UP-VLA-phi-w/o-mmu*	Prediction	ABC→D	0.844	0.705	0.604	0.520	0.430	3.13
UP-VLA	Prediction&VLA	ABC→D	0.928	**0.865**	**0.815**	**0.769**	**0.699**	**4.08**

Table 1. Zero-shot long-horizon evaluation on the Calvin benchmark where agent is asked to complete five chained tasks sequentially. Results marked with an asterisk (*) indicate those that we reproduced, while the rest are copied from the original papers. UP-VLA achieves the best performance, demonstrating that our approach exhibits strong generalization capabilities in simulated environments.

AI-Generated

tered scenes.

ZZZMARKtextbfRepresentation choice. Discrete future prediction via VQ-GAN codes proved more effective for conditioning actions than direct pixel regression or latent diffusion in our pilots. The code grid offers a compact spatial scaffold that preserves object boundaries and contact geometry while filtering out lighting and texture variations. This sharpens cross-attention between predicted futures and action tokens and stabilizes behavior across domains. Consistent with Table 3, Bridge pretraining is crucial; codebook dropout during pretraining and low-temperature top-k sampling at test time further reduced drift in clutter without noticeable latency.

ZZZMARKtextbfLatency and deployment. End-to-end inference maintains real-time budgets in our tests; batching language-only steps reduces overhead. PRE can be disabled adaptively when entropy is high or motion is small, trading a minor loss in foresight for lower compute.

7. Conclusion

We introduced UP-VLA, a unified autoregressive VLA that combines multi-modal understanding with future visual prediction. Through a task-aware prompting and attention scheme and a two-stage training recipe, UP-VLA captures both semantic grounding and low-level spatial dynamics. The approach achieves state-of-the-art results on CALVIN long-horizon benchmarks and improves real-world success, particularly on precise and unseen-object tasks. Ablations

Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABCD→D	0.844	0.617	0.438	0.323	0.227	2.45
Robo-Flamingo	VLA	ABCD→D	**0.964**	0.896	0.824	0.740	0.660	4.09
GR-1	Prediction	ABCD→D	0.949	0.896	0.844	0.789	0.731	4.21
UP-VLA	Prediction&VLA	ABCD→D	0.962	**0.921**	**0.879**	**0.842**	**0.812**	**4.42**

Table 2. Zero-shot long-horizon evaluation on the Calvin ABCD→D benchmark. Results of baselines are copied from original papers.

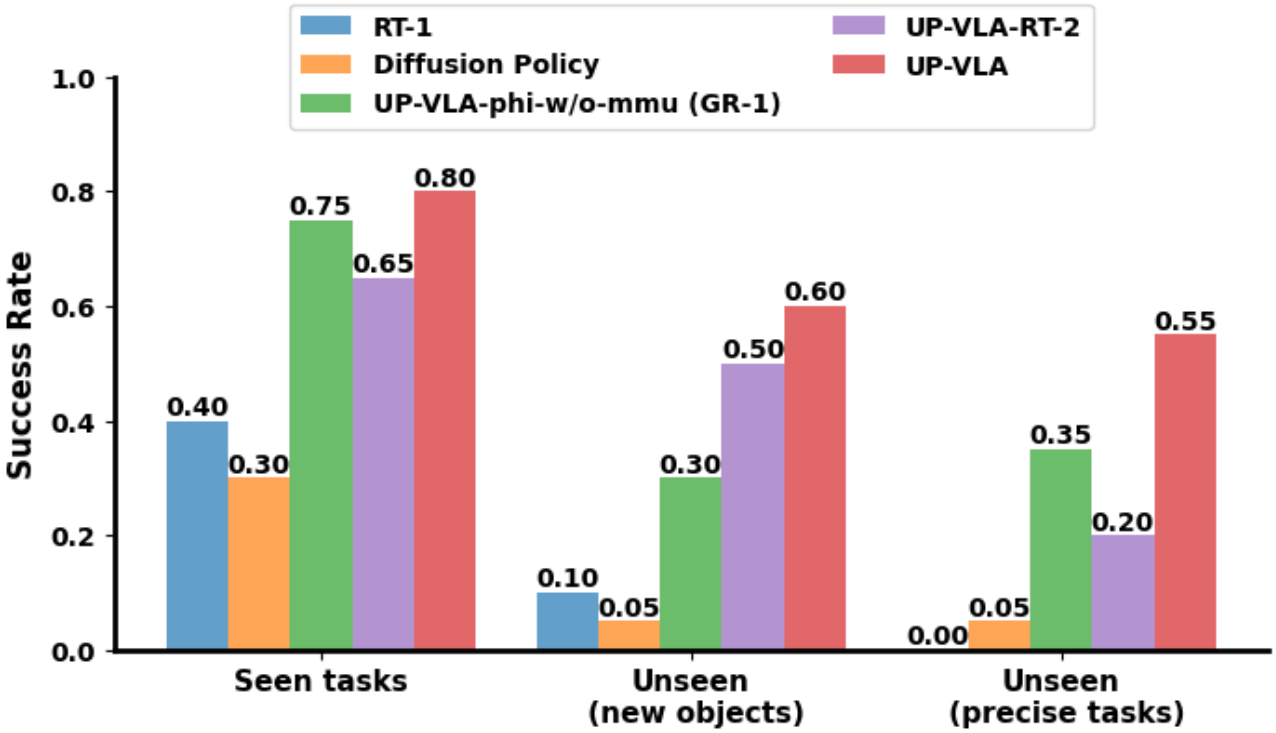


Figure 7. Real-world manipulation results.

Method	ABC→D	Real World	
	Avg.Len↑	Seen↑	Unseen↑
w/o MMU	3.89	0.85	0.20
w/o Bridge-Pretrain	2.74	0.65	0.30
w/o Prediction	1.44	0.65	0.35
w/o MMU-Condition	3.99	0.80	0.50
Full	4.08	0.80	0.58

Table 3. Ablating components of UP-VLA.

confirm the complementary roles of prediction and understanding in achieving these gains.

References

Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., Gopalakrishnan, K., Hausman, K., Herzog, A., Hsu, J., et al. Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022.

Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Chen, X., Choromanski, K., Ding, T., Driess, D., Dubey, A., Finn, C., et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

Chi, C., Feng, S., Du, Y., Xu, Z., Cousineau, E., Burchfiel, B., and Song, S. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

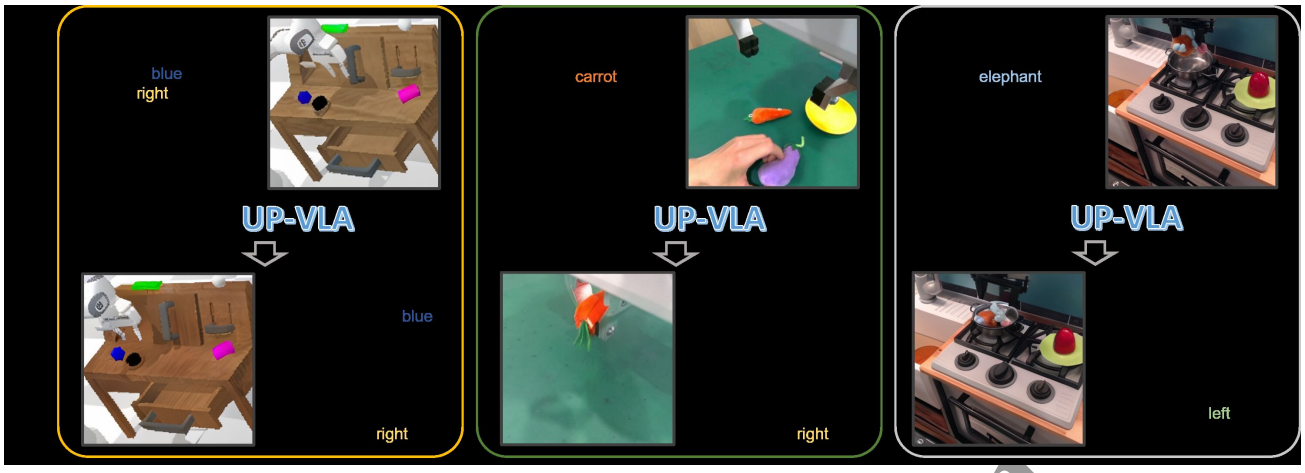


Figure 8. Qualitative examples of MMU outputs and PRE future predictions.

AI-Generated

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

Esser, P., Rombach, R., and Ommer, B. Taming transformers for high-resolution image synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 12873–12883, 2021.

Ke, T.-W., Gkanatsios, N., and Fragkiadaki, K. 3d diffuser actor: Policy diffusion with 3d scene representations. *arXiv preprint arXiv:2402.10885*, 2024.

Kim, M., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., Vuong, Q., Kollar, T., Burchfiel, B., Tedrake, R., Sadigh, D., Levine, S., Liang, P., and Finn, C. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

Li, X., Liu, M., Zhang, H., Yu, C., Xu, J., Wu, H., Cheang, C., Jing, Y., Zhang, W., Liu, H., et al. Vision-language foundation models as effective robot imitators. *arXiv preprint arXiv:2311.01378*, 2023a.

Li, Y., Bubeck, S., Eldan, R., Del Giorno, A., Gunasekar, S., and Lee, Y. T. Textbooks are all you need ii: phi-1.5 technical report. *arXiv preprint arXiv:2309.05463*, 2023b.

Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in neural information processing systems*, 36, 2024.

Mees, O., Hermann, L., Rosete-Beas, E., and Burgard, W. Calvin: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks. *IEEE Robotics and Automation Letters (RA-L)*, 7(3):7327–7334, 2022.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.

Walke, H., Black, K., Lee, A., Kim, M. J., Du, M., Zheng, C., Zhao, T., Hansen-Estruch, P., Vuong, Q., He, A., Myers, V., Fang, K., Finn, C., and Levine, S. Bridgedata v2: A dataset for robot learning at scale. In *Conference on Robot Learning (CoRL)*, 2023.

Wu, H., Jing, Y., Cheang, C., Chen, G., Xu, J., Li, X., Liu, M., Li, H., and Kong, T. Unleashing large-scale video generative pre-training for visual robot manipulation. *arXiv preprint arXiv:2312.13139*, 2023.

Zhen, H., Qiu, X., Chen, P., Yang, J., Yan, X., Du, Y., Hong, Y., and Gan, C. 3d-vla: A 3d vision-language-action generative world model. *arXiv preprint arXiv:2403.09631*, 2024.